

Jiawei Chen

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*Only a profound understanding of the underlying physics of turbulence can pave the way for
truly effective turbulence modeling*

Last updated 07/24/2026

EDUCATION

Ph.D. in Mechanics

Tsinghua University, Beijing, China (2024)

M.S.E. in Power Engineering

Harbin Institute of Technology, Harbin, China (2019)

B.S.E. in Energy and Power Engineering

Harbin Engineering University, Harbin, China (2017)

PROFESSIONAL EXPERIENCE

Iowa State University

Postdoctoral Research Associate 2025 – Present

University of Texas at Arlington

Postdoctoral Research Associate 2024 – 2025

RESEARCH INTERESTS

- Fluid Mechanics
- Multiphase Flows and Heat Transfer
- Agentic AI & Large Language Models
- Generative Learning for Fluid Dynamics

RESEARCH PROJECTS

- An Integrated Numerical and Experimental Study of Wind-Driven Water Film Flow Dynamics Pertinent to Wind Turbine Icing Phenomena. National Science Foundation (NSF), Award AWD-029845-00001. Total award: \$620K. 2024–2027. PI: Dr. Ping He. My role: Postdoctoral Research Associate.
- Development and Application of Liutex and Third Generation of Vortex Definition and Identification. National Science Foundation (NSF) EAGER Award 2300052. PI: Dr. Chaoqun Liu. My role: Postdoctoral Research Associate, 2024–2025.

RESEARCH PROPOSAL & TRAINING

Research proposal: conceptualized, drafted and led one NSF proposal as PI

- A Large Language Model–Driven Agentic Framework for Automated Turbulence Model Development. National Science Foundation (NSF) Computational and Data-Enabled Science and Engineering (CDS&E). Total award: \$300K. 2027–2030. **PI: Dr. Jiawei Chen.** Co-PI: Dr. Ping He. (to be submitted)

Training

- **Completed independent grant-writing training.** Completed the Center for Communication Excellence (CCE) Grantsmanship Summer Intensive microcredential, Iowa State University (Badge Level 1: Finding Grants; Badge Level 2: Preparing Applications), covering funder identification, eligibility and competitiveness assessment, and proposal narrative development; produced a project Gantt chart and a response to a mock Funding Opportunity Announcement.
- **ISU-approved Principal Investigator, registered with NSF Research.gov.** Completed Iowa State University’s Research Security Training, modeled after NSF, NIH, DoD, and DoE guidelines (export controls, disclosure policies, risk mitigation, and responsible international collaboration).

OPEN-SOURCE SOFTWARE

PhysMiner: Agentic-AI framework for automated turbulence-physics discovery, integrating triple-decomposition flow analysis with LLM-driven reasoning

Status: Open to the public; validated end-to-end on periodic-hill flow, where it autonomously derived an improved subgrid-scale model with superior Reynolds-stress predictions

Reference: **Chen, J.,** Gao, H., & He, P. (2026). *Computer Methods in Applied Mechanics and Engineering*, under review

JAX-LES: Differentiable computational fluid dynamics solver built on JAX, enabling gradient-based optimization and machine-learning-augmented turbulence modeling directly within the flow solver

Status: In active development (work in progress); validated on decaying isotropic turbulence and turbulent channel flow; planned for open-source release

TEACHING AND MENTORING

Teaching Preparation

Preparing Future Faculty (PFF) Program, Iowa State University

Fall 2026

Selected participant; semester-long seminar (GRST 5850) covering course and job-material development, the teaching–research–service balance, and the promotion and tenure process, paired with biweekly meetings with a tenure-track faculty mentor.

Courses Prepared to Teach:

- *Core/Undergraduate:* Fluid Mechanics; Introduction to Aerodynamics; Numerical Methods in Aerospace/Mechanical Engineering
- *Graduate/Elective:* Computational Fluid Dynamics; Turbulence Modeling and Simulation; Agentic AI and Generative AI for Fluid Dynamics

Mentoring

Postdoctoral Research Associate

2024 – 2025

(Department of Mathematics, University of Texas at Arlington)

1. Mentored Emran Hossen (Ph.D. in Mathematics, graduated Spring 2025) on Computational Fluid Dynamics (CFD) and turbulence modeling for his doctoral dissertation.
2. Collaboratively worked on a Liutex-based subgrid model for Large Eddy Simulation (LES), leading to a publication and presentation at the AIAA SciTech 2025 Forum.
3. Mentee successfully secured a Postdoctoral Scholar position at Michigan Technological University.

PROFESSIONAL SERVICE

Journal Paper Reviewing

- *AIAA journal*
- *Physics of Fluids*
- *Aerospace Science and Technology*
- *International Journal of Heat and Fluid Flow*

Seminar/Conference Services

- Reviewer for GPPS-TC-2026 conference papers: GPPS Montreal2026, Montreal, Canada, September 9-11, 2026, <https://gpps.global/montreal26/>

PUBLICATIONS

Journal Papers Under Review

- [J3] **J. Chen**, H. Gao, and P. He. PhysMiner: An Agentic AI Framework for Discovering Turbulence Physics. *Computer Methods in Applied Mechanics and Engineering*, 2026. (Under Review) [arXiv:2607.04009](https://arxiv.org/abs/2607.04009).
- [J2] **J. Chen**, P. He, and H. Hu. Transformer-Diffusion Modeling of Unsteady Flow Over an Iced Airfoil. *AIAA journal*, 2026. (Under Review)
- [J1] **J. Chen**, Y. Yu, E. Hossen, and C. Liu. Near-Wall Scaling and Separation Prediction of a Rotation-Based Subgrid-Scale Stress Model. *Journal of Turbulence*, 2025. (Under Review) [arXiv:2507.20043](https://arxiv.org/abs/2507.20043).

Peer-Reviewed Journal Papers

- [J13] Z. Li, L. Fang, **J. Chen**, A. Sharma, and P. He. Field Inversion Machine Learning for Time-Resolved Unsteady Flows in Airfoil Dynamic Stall. *AIAA Journal*, 2026. (In Press)
- [J12] **J. Chen**, M. Xiao, Z. Zhou, and Y. Zhang. Numerical investigation of discontinuous ice effects on swept wings. *International Journal of Heat and Fluid Flow*, 120, p. 110391, 2026. [doi:10.1016/j.ijheatfluidflow.2026.110391](https://doi.org/10.1016/j.ijheatfluidflow.2026.110391)

- [J11] **J. Chen**, Z. Zhou, M. Xiao, and Y. Zhang. Flow-separation simulations of continuous and discontinuous ice on swept wings. *Aerospace Science and Technology*, 174, p. 111891, 2026. [doi:10.1016/j.ast.2026.111891](https://doi.org/10.1016/j.ast.2026.111891)
- [J10] Z. Zhou, M. Xiao, **J. Chen**, L. Li, and Y. Zhang. Numerical Investigation of Sweep Effects on Turbulent Flow over Iced Wings. *AIAA Journal*, 64(2), pp. 823–839, 2026. [doi:10.2514/1.J065820](https://doi.org/10.2514/1.J065820)
- [J9] M. Xiao, Y. Wang, F. Rodach, B. Font, M. Kurz, P. Suárez, D. Zhou, F. Alcántara-Ávila, T. Zhu, J. Liu, R. Montalà, **J. Chen**, J. Rabault, O. Lehmkuhl, A. Beck, J. Larsson, R. Vinuesa, and S. Pirozzoli. SmartFlow: A CFD-solver-agnostic deep reinforcement learning framework for computational fluid dynamics on HPC platforms. *Computer Physics Communications*, 2025. (In Press) [arXiv:2508.00645](https://arxiv.org/abs/2508.00645).
- [J8] **J. Chen**, C. Yan, and Y. Zhang. Separation Prediction and Roughness Effect Simulation of Iced Airfoils Based on a Modified $k\text{-}\omega\text{-}\gamma$ Transition Model. *Aerospace Science and Technology*, 162, p. 110202, 2025. [doi:10.1016/j.ast.2025.110202](https://doi.org/10.1016/j.ast.2025.110202)
- [J7] **J. Chen**, Y. Zhang, and S. Fu. Ice Accretion Simulation Incorporating Roughness Effects and Separation Correction through a Transition Model. *Physics of Fluids*, 36(4), 2024. [doi:10.1063/5.0209004](https://doi.org/10.1063/5.0209004) (Editor's Pick)
- [J6] **J. Chen**, Y. Zhang, and S. Fu. Numerical study of the effects of discontinuous ice on three-dimensional wings. *Physics of Fluids*, 36(7), 2024. [doi:10.1063/5.0217642](https://doi.org/10.1063/5.0217642)
- [J5] **J. Chen**, P. Yang, Y. Zhang, and S. Fu. Aerodynamic prediction and roughness implementation toward ice accretion simulation. *Physics of Fluids*, 36(1), 2024. [doi:10.1063/5.0184630](https://doi.org/10.1063/5.0184630)
- [J4] P. Yang, H. Li, **J. Chen**, Y. Duan, and Y. Zhang. Influence of Air Flow Field Computation on an Ice Accretion Simulation of Airfoils. *Aerospace Science and Technology*, 151, p. 109315, 2024. [doi:10.1016/j.ast.2024.109315](https://doi.org/10.1016/j.ast.2024.109315)
- [J3] **J. Chen**, Y. Zhang, and S. Fu. Roughness implementation and convective heat transfer coefficient computation toward ice accretion simulation. *Physics of Fluids*, 35(11), 2023. [doi:10.1063/5.0173847](https://doi.org/10.1063/5.0173847) (Featured Article)
- [J2] **J. Chen**, L. Cai, and S. Wang. Numerical Study on Effects of Different Excitation Forms on Forced Film Cooling Characteristics. *Journal of Propulsion Technology*, 41(4), pp. 860–867, 2020.
- [J1] S. Sun, S. Wang, S. Chen, C. Tao, L. Cai, and **J. Chen**. The impact of various forward sweep angles on the performance of an ultra-high-load low-reaction transonic compressor rotor. *Applied Thermal Engineering*, 150, pp. 953–966, 2019. [doi:10.1016/j.applthermaleng.2019.01.045](https://doi.org/10.1016/j.applthermaleng.2019.01.045)

Conference Papers

- [C3] **J. Chen**, Y. Yu, and C. Liu. A Relative Liutex Method for Vortex Identification. *Vortex 2025 Conference Proceedings*, 2025 (to appear online Fall 2026). [arXiv:2512.23857](https://arxiv.org/abs/2512.23857).

- [C2] **J. Chen**, Y. Yu, and C. Liu. A Dynamic Subgrid-Scale Model Based on Liutex Theory for Wall-Bounded Turbulent Flows. *Vortex 2025 Conference Proceedings*, 2025 (to appear online Fall 2026). [arXiv:2512.14953](https://arxiv.org/abs/2512.14953).
- [C1] E. Hossen, Y. Yu, C. Liu, **J. Chen**, Z. Liu, Y. Yan, V. Shah, and O. Alvarez. Liutex-Based Sub-Grid Model for Large Eddy Simulation of Turbulent Flow. *AIAA SciTech 2025 Forum*, AIAA 2025-0073, 2025. [doi:10.2514/6.2025-0073](https://doi.org/10.2514/6.2025-0073)

TECHNICAL SKILLS

Programming Languages

Matlab, Python, Fortran, C++, LaTeX

Software

OpenFOAM, ANSYS, SolidWorks, Paraview, Tecplot, Linux, Slurm

High-Performance Computing

- Texas Advanced Computing Center
- Nova cluster